Zuhao Zhang (张祖豪)

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Education

Shanghai Jiao Tong University | *M.S. student, John Hopcroft Center for Computer Science* **09/2025-03/2028**

- Supervisor: Prof. Shuai Li. Research directions: reinforcement learning, post-training for LLMs, agent evaluation, and AI for Science.

Tongji University | *B.Eng. in Computer Science and Technology* **09/2021-06/2025**

- GPA: 4.77/5.00; Rank: 21/130. Top Honors for Graduation Thesis (Top 0.2%).

Research Interests

Interests: Reinforcement Learning, LLM Post-training, Reward Modeling, LLM Reasoning, Agent Evaluation, AI for Science

My research focuses on reinforcement learning and post-training for large language models, with an emphasis on reward modeling, LLM reasoning, and agent evaluation. I am currently working on LLM-powered assistant evaluation and RLHF-style reward modeling, which together shape my interest in scalable feedback, evaluation, and alignment for foundation models.

Publications

- **Zuhao Zhang**, Yuyue Cheng, Yuante Li, Chengyi Zhuang, Linjian Mo, Shuai Li, "MiniAppBench: Evaluating the Shift from Text to Interactive HTML Responses in LLM-Powered Assistants." ICML 2026 **Spotlight** (Oral Pending).
- **Zuhao Zhang**, Zhijie Wang, Shuai Li, Meng Jin, Xinbing Wang, "ERiC: Entropic Risk-aware Inference-in-the-loop Control for Adaptive Seawall Planning." Submitted to UAI 2026.

Selected Research Projects

MiniAppBench / MiniAppEval | *LLM-powered assistant evaluation benchmark and evaluation framework* **11/2025-01/2026**

- Led the first benchmark for evaluating interactive HTML MiniApps generated by LLM-powered assistants, using large-scale real-world user queries and real-world task principles.
- Designed MiniAppEval, combining static code/Eval-Ref checks with Playwright-based dynamic interaction testing; achieved 90%+ human agreement in evaluation.
- Released the project homepage and open-source materials to support reproducible evaluation of customized, highly interactive assistant outputs.

ERiC | *Risk-aware inference-in-the-loop control for adaptive seawall planning* **07/2025-02/2026**

- Proposed a belief-space MPC framework that unifies online inference with risk-sensitive robust control under partial observability, tail disaster risk, and structural model uncertainty.
- Built an AR6/FACTS-aligned workflow-aware world model and amortized variational filter for online belief tracking of sea-level-rise observations; workflow posterior macro-F1 reached 0.89 with ECE around 0.035.
- In the NYC case study, reduced catastrophic-year risk under SSP5-8.5 to 2e-5, about 260x lower than a static planning baseline.

LLM-Based Scientific Discovery | *Evolutionary Math Agent for high-dimensional error-correcting codes* **12/2025-Present**

- Built a Math Agent framework for code self-evolution to search high-dimensional error-correcting codes and approach theoretical bounds.
- Found a [31, 16, 7] optimal linear code through code-evolution search; currently exploring still-unverified theoretical upper bounds.

Industry and Research Experience

Ant Group | *CTO-Emerald-Nova-Agent Research Intern, advised by Chenyi Zhuang* **10/2025-03/2026**

- Served as the project lead for MiniAppBench, producing the ICML 2026 Spotlight paper and building the open-source project presence.
- Built an internal reward server and evaluation pipeline adopted by internal model training and evaluation teams, including Nova/Lingguang-related workflows.

- Deeply participated in Lingguang evaluation, designing an Intention/Static/Dynamic evaluation taxonomy with Eval-Ref support; summarized model failure modes and proposed defect-driven benchmarks and template-based fixes that were adopted by the team.
- Contributed to the AWorld framework by fixing MCP tool-calling bugs in online high-concurrency environments, improving training and evaluation efficiency.

DiDi | *IBG Marketplace, Algorithm Transaction Strategy, advised by Zihao Lu* **03/2026-Present**

- Developed an RLHF-style reward modeling pipeline for personalized order-hold policy optimization in ride-hailing marketplaces.
- Modeled human reward signals and online feedback as optimizable reward targets, improving order-hold effectiveness compared with manually designed rewards while reducing reliance on exhaustive policy search.
- Validated and launched the strategy, improving dispatch success rate by 0.2%; the policy has been deployed in Mexico and Brazil.

Awards

- Top Honors for the Graduation Thesis (Top 0.2%) at Tongji University.
- Tongji University Academic Scholarship - First-class, 2024 (Top 5%).
- Tongji University Academic Scholarship - First-class, 2023 (Top 5%).
- Tongji University Academic Scholarship - First-class, 2022 (Top 5%).

Teaching Assistantships

- Advanced Language Program Design (Honor Class), Tongji University: Fall 2024, Fall 2023, Fall 2022; Spring 2024, Spring 2023.